ASSIGNMENT

TECHNOLOGY PARK MALAYSIA

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Temporal Consistency Problem in Diffusion-based Model : Multi Model Architecture Integrated with AnimateDiff and GNN

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Abstract – The entertainment industry is currently experiencing a rapid evolution due to high demands environment and introduce of artificial intelligent in this era. First phase of the research paper discusses existing problem and solution in video diffusion model then proposed a possible multi architecture design as the solution. The proposed system use Graph Neural Network(GNN) to solve limitation within the denoising technique and process. This research paper will be mapping how the literature review was conducted. It will provide justification and prove for proposed system. Planning and design for methodology which include target user, sampling method, data collection presented.

Key Terms: *AnimateDiff, diffusion model, entertainment industry, literature review map, multi model architecture, purposive sampling, temporal consistency*

1. Introduction

Driven by the high demands of the digital era and the emergence alert of artificial intelligence, the entertainment industry is undergoing a significant and rapid evolution. After analyzing difficulties and possible causes that cause the entire situation. The focus of first phase of this research paper analyzes the technical disadvantages of current video diffusion models alongside existing industry solutions. Building on this analysis, the study dive further into temporal inconsistency which causes instable video generation output. Next, the study proposes a multi-model architectural design as a potential framework which inspires from dynamic traffic system.

The following section will be introducing literature review of this research paper and methodology planned and designed to proposed. The literature review section will first provide a literature review map to show the literature review mapped. Corresponding elements will be explained and justify accordingly to show how the research questions and objectives are being discovered. Next, methodology section will present how this research paper is planned to collect feedback and evidence to support the research result using human study. Target user, sampling method, and data collection that is being chosen will be presented and justified accordingly.

2. Literature Review

2.1 Literature Review Map

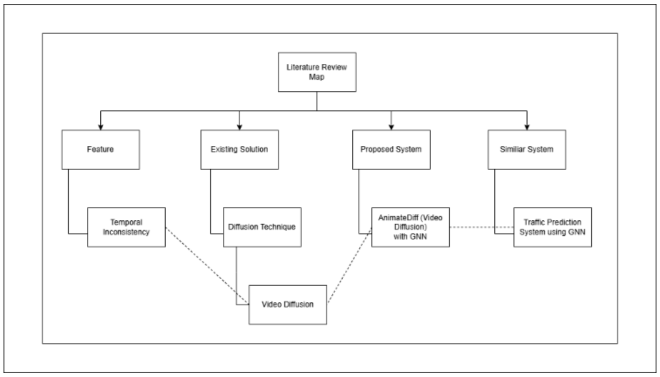

Figure 1: Literature Riview Map (Appendix 1)

2.2 Feature

The core technical challenge identified is the temporal consistency problem inherent in diffusion-based video generative models. In professional entertainment workflows, a failure to maintain stable visual attributes such as character identity, appearance, and motion results in flickering artifacts and unnatural, unstable motion trajectories. These issues are happen during long-term or multi-scene video generation because the models often lack stable long-range temporal constraints and global memory mechanisms which means that frames (He et al., 2023). Consequently, each generated frame may have limited similarity to previous ones, producing visually inconsistent results that are not suitable for professional industry standards.

2.3 Existing Solution

This research focuses on models like AnimateDiff, an open-source text-to-image models. Its strength is it is fully free and highly scalabe with different customization and extension tools. While AnimateDiff is highly flexible and compatible with extra tools like LoRA and ControlNet. it relies heavily on localized temporal modeling and customize enhancement (Guo et al., 2023). This localized approach causes it to struggle with maintaining frame-to-frame coherence over long sequences. Existing methods tried to attempt to solve this through improved sampling strategies or attention-based temporal modeling but maintaining character consistency across different frames remains an unresolved problem.

2.4 Proposed System

To resolve these inconsistencies, this research proposes a multi-model architecture that integrates Graph Neural Networks (GNNs) into the diffusion denoising process. The system represents a video sequence which is the frames as temporal graph format. In it frames acts as nodes and edges wich encode the relationship between them. Different from traditional attention-based methods, this architecture uses GNNs to perform message passing across the graph like messaging between multiple agents. This allows the model to refine latent representations by using contextual information from neighboring frames which will determine that information is transferred close across the entire sequence for a more stable and coherent video output.

2.5 Similar System

The design of this proposed system is specifically inspired by the success of temporal GNNs in modeling dynamic traffic systems. It use GNN to improve prediction between multiple roads with bunch of data in traffic flow prediction system which the overall structure is kinda similar to frames and denoising process in video diffusion. In this system, consistency over time is maintained through structured, relational interactions between different data points which will be representing relationship between frames in the proposed system. By transferring this principle to video diffusion, the research utilizes a dynamic adversarial graph convolutional network approach similar to that used for traffic flow to maintain structural relationships and reduce object drift (Chen et al., 2025). This cross-domain application of GNNs provides a robust framework for ensuring that visual elements remain consistent as the logical flow of the video progresses.

3.Methodology

3.1 Target User

Target User of the proposed system are content creators in entertaining industry. They are the direct user of the proposed system and the prompt provider for video generation. They will be integrating the proposed system into their existing workflow to enhance productivity of content creation. The system can be used for video content generation, film effect expansion, add missing frames in their workflow. The system usage can greatly decrease the content creation cost and time, especially while facing difficulty like actors displacement, limited cost and short production time.

3.2 Sampling Method

Purposive sampling is planned to be used as the sampling method for this research paper. This sampling methods is successfully implemented in other research paper that topic related to this research paper which is Digital “Double Hatters”: Augmenting Audiovisual Creative Work with a Generative Text-to-Video Workflow (Liu et al., 2025). Sampling targets are restricted to content creator that have experience of using video generator in their workflow. This is because they can ensure the prompt quality and provide more valuable insight or feedback for the proposed system. This research paper planned to have 20 professionals in entertainment industry do an interview to provide insights and feedback. 1000 online prototype testing and feedback using same questions as interview also planned to be conducted on social media platform in community of our target user to get more information.

3.3 Data Collection

Interview which is qualitative method is planned to be used for collecting target user’s feedback. The sample interview question is designed to have three section which is three demographic profile, four open-ended and four close-ended questions. Demographic profile section will gather interviewer professional backgrounds. Demographic profile section will be used to weight the feedback interviewers provided and confirm sampling targets. Closed ended problems will be used as evaluation for the system after interviewers finish prototype testing. Lastly, open ended question will collect qualitive feedback and suggestion from interviewers for future advancement and upgrade.

4. Conclusion

Literature review of this paper presented flow of idea for proposed system and proof of how the proposed system being justified. Literature review map provides a clearer view for structure of the entire literature review. Methodology part show planning for how to filter target user, sampling and collect data from sampling target for further analysis and feedback gathering. This research provide a meaningful insight of how temporal inconsistency can be solved. This paper also prove first section of the research about the connection between GNN and solving temporal inconsistency. Last, this paper planned how can the research be conducted in future and what methodology it might suitable.

Reference

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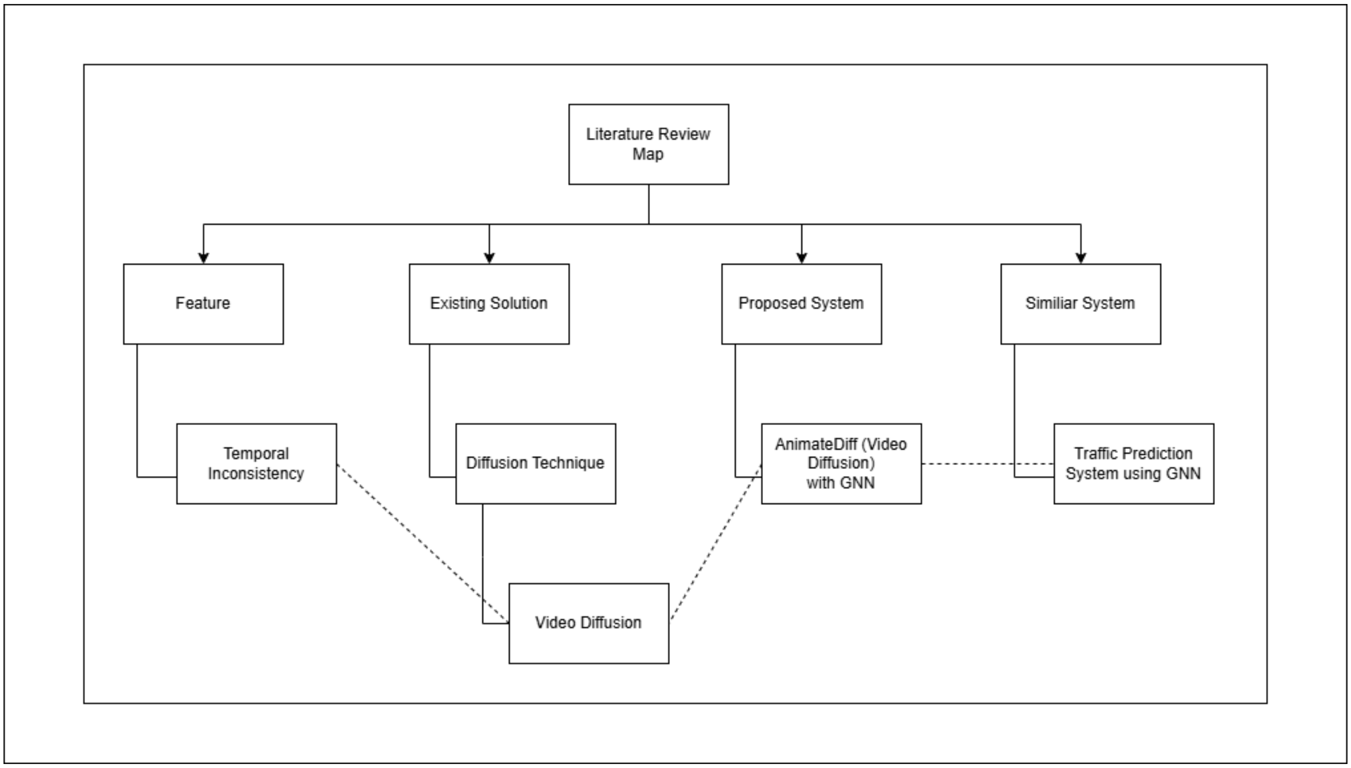
Appendix

1. Literature Review Map



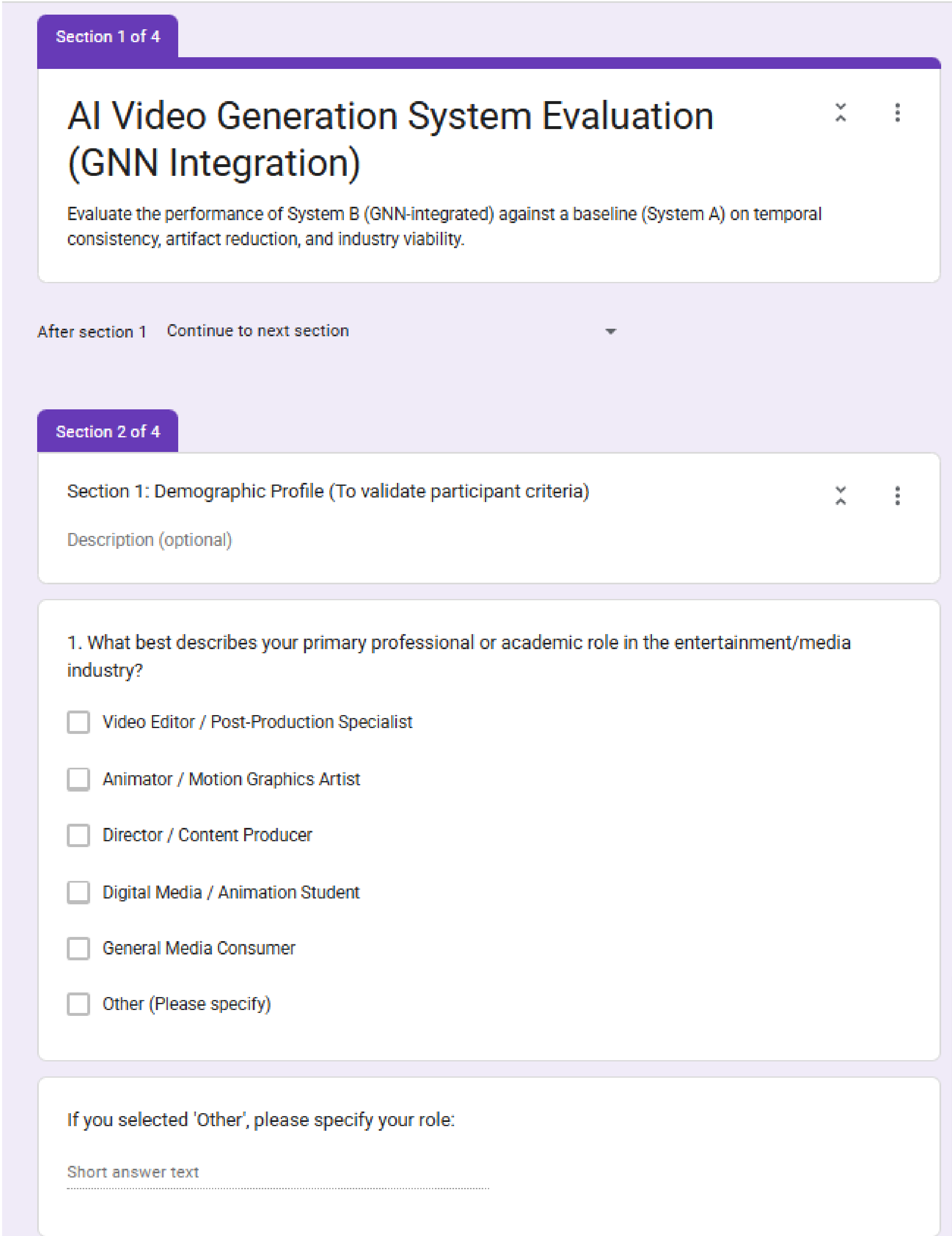
2. Similar System

Title	Author	Year	Solved Problem	Application	Industry
Temporal representation learning enhanced dynamic adversarial graph convolutional network for traffic flow prediction	Linlong Chen, Linbiao Chen, Hongyan Wang & Jian Zhao	2025	Temporal Inconsistency	Transport Prediction using Graph Neural Network	Transport

3. Interview Questions

Section 1: Demographic Profile (To validate participant criteria)
1. What best describes your primary professional or academic role in the entertainment/media industry?
2. How many years of experience do you have in digital video production or content creation?
3. Prior to this session, what was your familiarity with AI video generation tools (e.g., AnimateDiff, Runway, Sora)?
Section 2: Closed-Ended Questions (System Evaluation)
4. Temporal Smoothness: Based on the videos you just generated, how would you rate the natural flow of motion in System B (GNN-integrated) compared to System A (Baseline)?
5. Character/Object Identity Preservation: How effectively did System B maintain the exact visual identity of your prompted subject from the first frame to the final frame, compared to System A?
6. Reduction of Visual Artifacts: When evaluating the background and finer details (e.g., textures, edges), to what extent did System B successfully reduce flickering or warping compared to System A?
7. Commercial Viability: Considering the balance between the time it took to generate the video and the final visual quality, how viable is System B for professional entertainment content?
Section 3: Open-Ended Questions (Qualitative Feedback)
8. Prompt Responsiveness: How accurately did System B interpret your prompt's complex motions compared to System A, and did the added temporal consistency limit the creativity of the output?
9. Observation of Artifacts: While watching your generated videos side-by-side, what specific structural deficiencies (e.g., missing limbs, background warping, sudden color shifts) did you notice, and how did the GNN integration affect them?
10. Workflow Impact: If a tool like System B could consistently guarantee stable video generation without the usual flickering, how would that change the way you balance production costs and project timelines?
11. Final Thoughts: Do you have any additional feedback regarding the user experience of generating these videos, or suggestions for improving the visual output of the temporal graph framework?

4. Google Form Screenshot



Section 1 of 4

AI Video Generation System Evaluation (GNN Integration)

Evaluate the performance of System B (GNN-integrated) against a baseline (System A) on temporal consistency, artifact reduction, and industry viability.

After section 1 Continue to next section

Section 2 of 4

Section 1: Demographic Profile (To validate participant criteria)

Description (optional)

1. What best describes your primary professional or academic role in the entertainment/media industry?

- [ ] Video Editor / Post-Production Specialist
- [ ] Animator / Motion Graphics Artist
- [ ] Director / Content Producer
- [ ] Digital Media / Animation Student
- [ ] General Media Consumer
- [ ] Other (Please specify)

If you selected 'Other', please specify your role:

Short answer text

2. How many years of experience do you have in digital video production or content creation?

- ◯ Less than 1 year
- ◯ 1–3 years
- ◯ 4–7 years
- ◯ 8+ years

3. Prior to this session, what was your familiarity with AI video generation tools (e.g., AnimateDiff, Runway, Sora)?

- ◯ Expert (I use them regularly in my workflow)
- ◯ Intermediate (I have experimented with them occasionally)
- ◯ Beginner (I know of them but rarely use them)
- ◯ None (This is my first time using one)

Section 2: Closed-Ended Questions (System Evaluation)

4. Temporal Smoothness: Based on the videos you just generated, how would you rate the natural flow of motion in System B (GNN-integrated) compared to System A (Baseline)?

	1	2	3	4	5	
System B is significantly more choppy and disjointed	○	○	○	○	○	System B is significantly smoother and more natural

5. Character/Object Identity Preservation: How effectively did System B maintain the exact visual identity of your prompted subject from the first frame to the final frame, compared to System A?

	1	2	3	4	5	
Severe identity drift (the subject changed completely)	○	○	○	○	○	Perfect preservation of identity throughout the video

6. Reduction of Visual Artifacts: When evaluating the background and finer details (e.g., textures, edges), to what extent did System B successfully reduce flickering or warping compared to System A?

	1	2	3	4	5	
System B had much more flickering	○	○	○	○	○	System B completely eliminated flickering

7. Commercial Viability: Considering the balance between the time it took to generate the video and the final visual quality, how viable is System B for professional entertainment content?

	1	2	3	4	5	
Completely unviable (requires too much manual fixing)	○	○	○	○	○	Ready for immediate industry integration

Section 3: Open-Ended Questions (Qualitative Feedback)

8. Prompt Responsiveness: How accurately did System B interpret your prompt's complex motions compared to System A, and did the added temporal consistency limit the creativity of the output?

Your answer

9. Observation of Artifacts: While watching your generated videos side-by-side, what specific structural deficiencies (e.g., missing limbs, background warping, sudden color shifts) did you notice, and how did the GNN integration affect them?

Your answer

10. Workflow Impact: If a tool like System B could consistently guarantee stable video generation without the usual flickering, how would that change the way you balance production costs and project timelines?

Your answer

11. Final Thoughts: Do you have any additional feedback regarding the user experience of generating these videos, or suggestions for improving the visual output of the temporal graph framework?

Your answer

5. Google Form
   https://docs.google.com/forms/d/e/1FAIpQLScMiDBy7go7oi2ZK482O8kHq_emOQYRurMx7v79M70Mqu3zEA/viewform?usp=publish-editor